

Prateek Saha

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SUMMARY

I build full applications and the data and ML pieces that go inside them, mostly with Python, FastAPI and React. I finished my BCA in 2025 and am now doing an MCA, and my projects range from a deployed RAG study assistant to YOLOv8 vehicle detection. I also spent several months on 3D LiDAR annotation and QA, which taught me to keep accuracy high on large volumes of data. I am looking for a fresher role in full-stack development or data science.

SKILLS

Languages: Python, SQL, JavaScript, C++, TypeScript (basic), Java (foundational)

Backend: FastAPI, REST APIs, JWT authentication, Node.js, Pydantic

Frontend: React, Tailwind CSS, HTML, CSS

Data and ML: Pandas, NumPy, Scikit-learn, Keras, TensorFlow, PyTorch, YOLOv8, OpenCV, FAISS, RAG, EDA, Power BI, MS Excel

Databases and Cloud: Supabase, PostgreSQL, MySQL, Vercel, Render

Tools: Git, GitHub, VS Code, Jupyter Notebook, Unit testing in Python

PROJECTS

AI Study Companion

React, FastAPI, Supabase, FAISS, Cohere

Live: rag-v2-gules.vercel.app

- Built a study assistant that answers questions from notes a user uploads, using retrieval-augmented generation.
- Gave each user their own FAISS index, with Cohere embeddings used to pull the right context before the model answers.
- Added quiz generation, weak-topic detection, adaptive roadmaps and a learning analytics dashboard.
- Secured the API with JWT and Supabase Row Level Security, and deployed on Vercel and Render.

MarketFlip

React, FastAPI, Supabase, PostgreSQL

Live: marketflip-mauve.vercel.app

- Developing a reverse marketplace where buyers post what they need and nearby shop owners bid on it.
- Built the request, bid and finalize flow, with seller contact details revealed only after a bid is finalized.
- Frontend on Vercel, FastAPI backend on Render, auth through Supabase Auth.

Sentinel.v8, Vehicle Detection

Python, YOLOv8, Streamlit, OpenCV

- Trained YOLOv8 on a custom set of about 979 images from IDD, Bengaluru and Bhilai covering cars, bikes and trucks.
- Converted XML annotations to YOLO format and compared training runs of 10, 25 and 30 epochs.
- Built a Streamlit app for image and video inference. Latest run: precision about 0.81, mAP@50 about 0.63.

Loan Approval Prediction

Python, Scikit-learn, Pandas, Seaborn

- Compared KNN, Random Forest, SVM and Logistic Regression on a 13-feature loan dataset after cleaning, EDA and encoding.
- Random Forest scored best at 98.32% accuracy.

Sales Analysis Dashboard (AdventureWorks)

MySQL, Power BI, DAX, Power Query

- Pulled data with SQL, cleaned it in Power Query and modeled it as a star schema.
- Wrote DAX measures for sales, profit margin, variance against budget and customer ranking, and built an interactive dashboard with slicers.

EXPERIENCE

Process Executive Intern, AugTech NextWealth IT Services, Bhilai

June 2025 to January 2026

- Annotated and validated 3D LiDAR data in GT Studio, covering vehicles, traffic signals, attributes and occlusion.
- Kept accuracy at 95%+ across all assigned work while meeting productivity targets.
- Did QA checks on annotated data and supported client queries when needed.

EDUCATION

MCA, Swami Vivekanand Technical University

2025 to 2027 (pursuing)

BCA, Hemchand Yadav Vishwavidyalaya, Durg (Kalyan PG College, Bhilai)

2021 to 2025

CERTIFICATIONS

- Web Development Bootcamp, Udemy (Colt Steele), 2025
- Machine Learning and Data Science, Udemy (Kirill Eremenko, Hadelin de Ponteves), ongoing
- Learn SQL using MySQL and Database Design, Scaler, 2024
- Data Structures and Algorithms in C++, Scaler, 2024